

EECS 16A Foundations of Signals, Systems, & Information Processing

Fall 2025 Homework 8

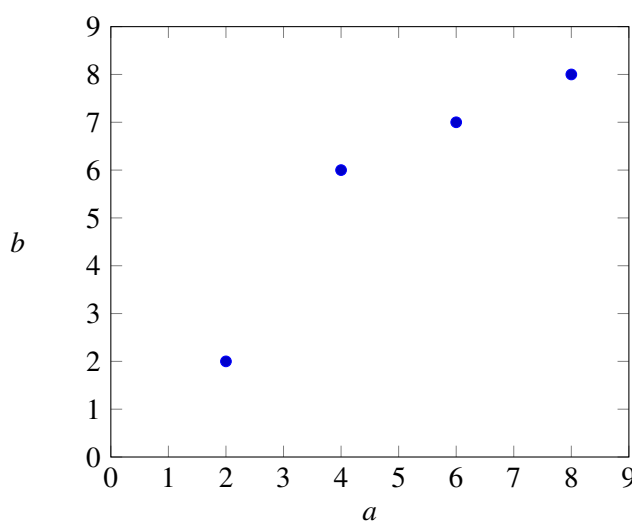
This homework is due Friday, October 24, 2025 at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Mechanical: Linear Least Squares



(a) Consider the above data points. Find the linear model of the form

$$\mathbf{b} = \mathbf{a}x$$

that best fits the data, i.e. find the scalar value of $x = \hat{x}$ that minimizes the squared error

$$\|\mathbf{e}\|^2 = \|\mathbf{b} - \mathbf{a}x\|^2 = \left\| \begin{bmatrix} b_1 \\ \vdots \\ b_4 \end{bmatrix} - \begin{bmatrix} a_1 \\ \vdots \\ a_4 \end{bmatrix} x \right\|^2. \quad (1)$$

Note: By using this linear model, we are implicitly forcing the fit equation to go through the origin.

Do not use IPython for this calculation and show your work. Once you’ve computed $\hat{x}$, compute the squared error between your model’s prediction and the actual $\mathbf{b}$ values as shown in Equation 1.

Plot the best fit line along with the data points to examine the quality of the fit. (It is okay if your plot of $\mathbf{b} = \mathbf{a}x$ is approximate.)

Solution:

Define $\mathbf{a} = [2 \ 4 \ 6 \ 8]^T$ and $\mathbf{b} = [2 \ 6 \ 7 \ 8]^T$. Applying the linear least squares formula, we get

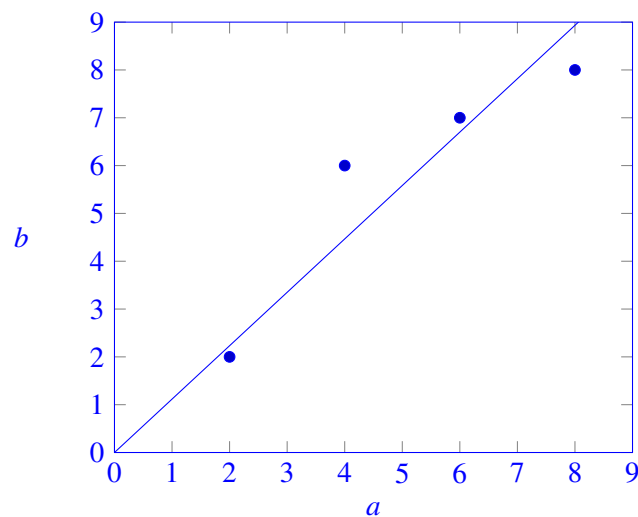
$$\begin{aligned}\hat{x} &= (\mathbf{a}^T \mathbf{a})^{-1} \mathbf{a}^T \mathbf{b} \\ &= \left([2 \ 4 \ 6 \ 8] \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix} \right)^{-1} [2 \ 4 \ 6 \ 8] \begin{bmatrix} 2 \\ 6 \\ 7 \\ 8 \end{bmatrix} \\ &= (120)^{-1} (134) = 1.1167\end{aligned}$$

The error between the model's prediction and actual b values is

$$\begin{aligned}\mathbf{e} &= \mathbf{b} - \hat{\mathbf{b}} = \mathbf{b} - \hat{x}\mathbf{a} \\ &= \begin{bmatrix} 2 \\ 6 \\ 7 \\ 8 \end{bmatrix} - 1.1167 \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} -0.234 \\ 1.534 \\ 0.3 \\ -0.934 \end{bmatrix}\end{aligned}$$

and the sum of squared errors is

$$\mathbf{e}^T \mathbf{e} = 3.367$$



- (b) You will notice from your graph that you can get a better fit by adding a b -intercept. That is we can get a better fit for the data by assuming a linear model of the form

$$\mathbf{b} = x_1 \mathbf{a} + x_2.$$

In order to do this, we need to augment our $\mathbf{A}$ matrix for the least squares calculation with a column of 1's (do you see why?), so that it has the form

$$\mathbf{A} = \begin{bmatrix} a_1 & 1 \\ \vdots & \vdots \\ a_4 & 1 \end{bmatrix}.$$

Find x_1 and x_2 that minimize the squared error

$$\|\mathbf{e}\|^2 = \|\mathbf{b} - \mathbf{Ax}\|^2 = \left\| \begin{bmatrix} b_1 \\ \vdots \\ b_4 \end{bmatrix} - \begin{bmatrix} a_1 & 1 \\ \vdots & \vdots \\ a_4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2. \quad (2)$$

Do not use IPython for this calculation and show your work.

Compute the squared error between your model's prediction and the actual $\mathbf{b}$ values as shown in Equation 2. Plot your new linear model. Is it a better fit for the data?

Solution:

Let $\mathbf{x} = [x_1 \ x_2]^T$. Using the linear least squares formula with the new augmented $\mathbf{A}$ matrix, we calculate the optimal approximation of $\mathbf{x}$ as

$$\begin{aligned} \hat{\mathbf{x}} &= (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} \\ &= \left(\begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 1 \\ 6 & 1 \\ 8 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 7 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} 120 & 20 \\ 20 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 7 \\ 8 \end{bmatrix} \\ &= \frac{1}{120(4) - 20(20)} \begin{bmatrix} 4 & -20 \\ -20 & 120 \end{bmatrix} \begin{bmatrix} 134 \\ 23 \end{bmatrix} \\ \hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} &= \begin{bmatrix} 0.95 \\ 1 \end{bmatrix} \end{aligned}$$

The linear model's prediction of $\mathbf{b}$ is given by

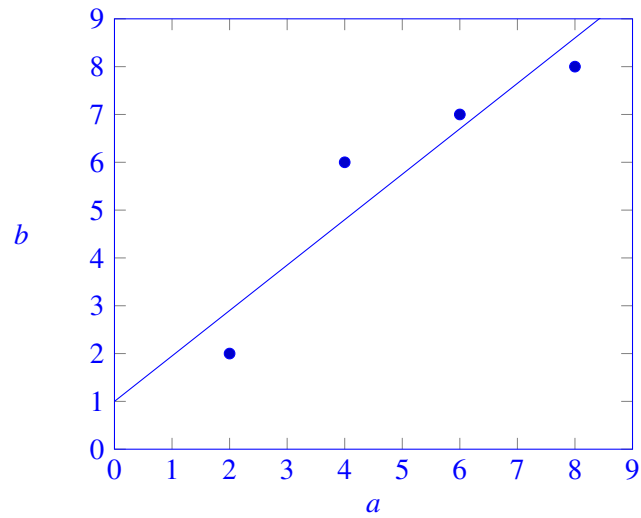
$$\hat{\mathbf{b}} = \mathbf{A}\hat{\mathbf{x}} = \begin{bmatrix} 2 & 1 \\ 4 & 1 \\ 6 & 1 \\ 8 & 1 \end{bmatrix} \begin{bmatrix} 0.95 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.9 \\ 4.8 \\ 6.7 \\ 8.6 \end{bmatrix}$$

and the error is given by

$$\mathbf{e} = \mathbf{b} - \hat{\mathbf{b}} = [-0.9 \ 1.2 \ 0.3 \ -0.6]^T$$

The summed squared error is

$$\|\mathbf{e}\|^2 = \mathbf{e}^T \mathbf{e} = 2.7$$



We can see both qualitatively from the plots and quantitatively from the sum of the squared errors that the fit is better with the b -intercept.

2. Study Spaces: Grimes Engineering Center vs Doe Library

A campus is tracking where students choose to study each evening. At each timestep n , every student studies at one of two locations:

- Grimes Engineering Center ($x_{\text{Grimes}}[n]$),
- Doe Library ($x_{\text{Doe}}[n]$).

We define the state vector:

$$\mathbf{x}[n] = \begin{bmatrix} x_{\text{Grimes}}[n] \\ x_{\text{Doe}}[n] \end{bmatrix}.$$

Observed switching behavior:

- 70% of students at Grimes return to Grimes next time; 30% switch to Doe.
- 60% of students at Doe return to Doe next time; 40% switch to Grimes.

Thus,

$$\mathbf{x}[n+1] = A\mathbf{x}[n] + \mathbf{u}[n], \quad A = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix}.$$

External input (renovations). Grimes Engineering Center has just completed major renovations, drawing additional new students beyond the usual returners. Each timestep during the promotion:

$$\mathbf{u}[n] = \begin{cases} \begin{bmatrix} 50 \\ 20 \end{bmatrix}, & n = 0, 1, \\ \mathbf{0}, & n \geq 2, \end{cases}$$

where the 50 represents new students directly attracted to Grimes because of the renovations, while the 20 represents extra students who also choose Doe because overall study activity on campus increases.

The initial state is $\mathbf{x}[0] = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$.

(a) Compute $\mathbf{x}[1]$, $\mathbf{x}[2]$, $\mathbf{x}[3]$.

Solution: Step $0 \rightarrow 1$:

$$A\mathbf{x}[0] = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 100 \\ 50 \end{bmatrix} = \begin{bmatrix} 90 \\ 60 \end{bmatrix}, \quad \mathbf{x}[1] = \begin{bmatrix} 90 \\ 60 \end{bmatrix} + \begin{bmatrix} 50 \\ 20 \end{bmatrix} = \begin{bmatrix} 140 \\ 80 \end{bmatrix}.$$

Step $1 \rightarrow 2$:

$$A\mathbf{x}[1] = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 140 \\ 80 \end{bmatrix} = \begin{bmatrix} 130 \\ 90 \end{bmatrix}, \quad \mathbf{x}[2] = \begin{bmatrix} 130 \\ 90 \end{bmatrix} + \begin{bmatrix} 50 \\ 20 \end{bmatrix} = \begin{bmatrix} 180 \\ 110 \end{bmatrix}.$$

Step $2 \rightarrow 3$ (no input):

$$\mathbf{x}[3] = A\mathbf{x}[2] = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 180 \\ 110 \end{bmatrix} = \begin{bmatrix} 170 \\ 120 \end{bmatrix}.$$

- (b) Let the total number of students be $T[n] = x_{\text{Grimes}}[n] + x_{\text{Doe}}[n] = \mathbf{1}^\top \mathbf{x}[n]$ with $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Show that

$$T[n+1] = T[n] + \mathbf{1}^\top \mathbf{u}[n],$$

and interpret this result.

Hint: Do the columns of A sum to 1?

Solution: Each column of A sums to 1, so $\mathbf{1}^\top A = \mathbf{1}^\top$. Then

$$T[n+1] = \mathbf{1}^\top \mathbf{x}[n+1] = \mathbf{1}^\top (A\mathbf{x}[n] + \mathbf{u}[n]) = \mathbf{1}^\top \mathbf{x}[n] + \mathbf{1}^\top \mathbf{u}[n] = T[n] + \mathbf{1}^\top \mathbf{u}[n].$$

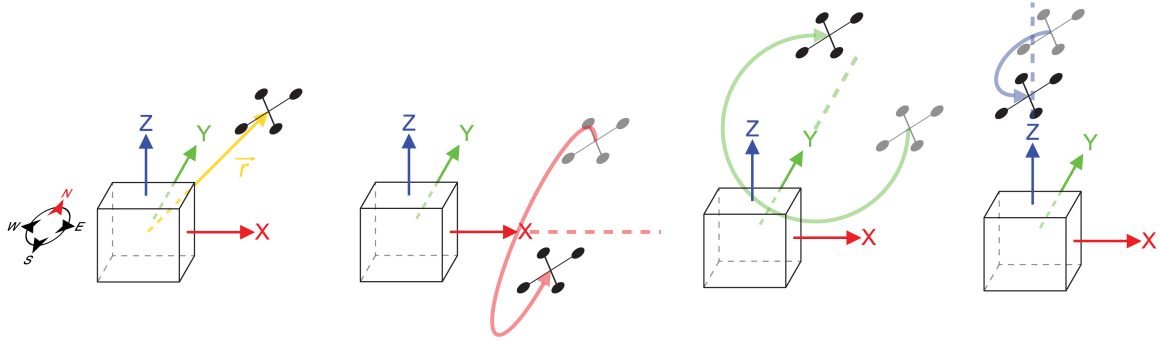
Interpretation: without renovations ($\mathbf{u}[n] = 0$), the total number of students is conserved—they only switch locations. Renovations add directly to the total.

- (c) Suppose the renovations continue indefinitely with $\mathbf{u}[n] = \begin{bmatrix} 50 \\ 20 \end{bmatrix} \forall n$. Without eigenvalues, describe qualitatively what happens as $n \rightarrow \infty$.

Solution: Each step adds $50 + 20 = 70$ new students, so $T[n]$ increases by about 70 each time. The matrix A just shifts students back and forth between Grimes and Doe. Over time the proportions at each location settle to a steady ratio, but the total number of students keeps growing without limit because new people keep arriving every step.

3. Quadcopter Transformations

Sanjay and his colleagues are interested in developing a communication link using a laser to control the location of a quadrotor. Consider a vector $\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$ representing the location of the quadcopter relative to the origin. The quadcopter is only capable of three different maneuvers relative to the origin. The maneuvers are rotations about the x , y , and z axes. For perspective, the positive x -axis points east, the positive y -axis points north, and the positive z -axis points up towards the sky. The figures below illustrate the quadcopter and these maneuvers.



We can represent each of these rotations, that are linear transformations, as matrices that operate on the location vector of the quadcopter, $\mathbf{r}$, to position it at its new location. The matrices $\mathbf{R}_x(\theta)$, $\mathbf{R}_y(\psi)$, and $\mathbf{R}_z(\phi)$ represent rotations about the x -axis, y -axis, and z -axis, respectively. The matrices are:

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}, \mathbf{R}_y(\psi) = \begin{bmatrix} \cos \psi & 0 & \sin \psi \\ 0 & 1 & 0 \\ -\sin \psi & 0 & \cos \psi \end{bmatrix}, \mathbf{R}_z(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- (a) Sanjay wants to make the quadcopter rotate first by $\frac{\pi}{6}$ about the x -axis, and then by $\frac{\pi}{3}$ about the z -axis. Use $\mathbf{R}_x(\theta)$, $\mathbf{R}_y(\psi)$, and $\mathbf{R}_z(\phi)$ to construct **a single matrix that performs the operations in the specified order**. Show the matrix operations and calculations by hand.

Solution: First, we must apply $\mathbf{R}_x(\theta)$ to $\mathbf{r}$, then $\mathbf{R}_z(\phi)$. We are told the angle of rotation about x -axis, θ , is $\frac{\pi}{6}$. Likewise, for the z -axis, $\phi = \frac{\pi}{3}$. So the matrix is:

$$\begin{aligned} \mathbf{R}_z\left(\frac{\pi}{3}\right) \mathbf{R}_x\left(\frac{\pi}{6}\right) &= \begin{bmatrix} \cos \frac{\pi}{3} & -\sin \frac{\pi}{3} & 0 \\ \sin \frac{\pi}{3} & \cos \frac{\pi}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{6} & -\sin \frac{\pi}{6} \\ 0 & \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} \cdot 1 + 0 + 0 & 0 - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + 0 & 0 + -\frac{\sqrt{3}}{2} \cdot (-\frac{1}{2}) + 0 \\ \frac{\sqrt{3}}{2} \cdot 1 + 0 + 0 & 0 + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} + 0 & 0 + \frac{1}{2} \cdot (-\frac{1}{2}) + 0 \\ 0 + 0 + 0 & 0 + 0 + 1 \cdot \frac{1}{2} & 0 + 0 + 1 \cdot \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{4} & -\frac{1}{4} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \end{aligned}$$

- (b) Sanjay accidentally punched in the two rotation commands in reverse. The $\frac{\pi}{3}$ rotation about the z -axis occurred before the $\frac{\pi}{6}$ rotation about the x -axis. Use $\mathbf{R}_x(\theta)$, $\mathbf{R}_y(\psi)$, and $\mathbf{R}_z(\phi)$ to construct a **single matrix that performs the operations in the accidentally reversed order**. Show the matrix operations and calculations by hand.

Solution: Now, we compute the matrix-matrix product with the order of $\mathbf{R}_z(\phi)$ and $\mathbf{R}_x(\theta)$ reversed.

$$\begin{aligned}\mathbf{R}_x\left(\frac{\pi}{6}\right)\mathbf{R}_z\left(\frac{\pi}{3}\right) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\frac{\pi}{6} & -\sin\frac{\pi}{6} \\ 0 & \sin\frac{\pi}{6} & \cos\frac{\pi}{6} \end{bmatrix} \begin{bmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3} & 0 \\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \cdot \frac{1}{2} + 0 + 0 & 1 \cdot \left(-\frac{\sqrt{3}}{2}\right) + 0 + 0 & 0 + 0 + 0 \\ 0 + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + 0 & 0 + \frac{\sqrt{3}}{2} \cdot \frac{1}{2} + 0 & 0 + 0 - \frac{1}{2} \cdot 1 \\ 0 + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} + 0 & 0 + \frac{1}{2} \cdot \frac{1}{2} + 0 & 0 + 0 + \frac{\sqrt{3}}{2} \cdot 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{3}{4} & \frac{\sqrt{3}}{4} & -\frac{1}{2} \\ \frac{\sqrt{3}}{4} & \frac{1}{4} & \frac{\sqrt{3}}{2} \end{bmatrix}\end{aligned}$$

- (c) Say the quadcopter was initially positioned at $\mathbf{r} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

- Where did Sanjay intend for the quadcopter to end up? Use your result from part (a) to find this out.
- Where did the quadcopter actually end up with the accidentally reversed order of the rotations? Use your result from part (b) to find this out.
- Did the quadcopter end up where it was supposed to go?

Solution: The intended location is given by:

$$\mathbf{R}_z\left(\frac{\pi}{3}\right)\mathbf{R}_x\left(\frac{\pi}{6}\right)\mathbf{r} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{4} & -\frac{1}{4} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{-1+2\sqrt{3}}{4} \\ \frac{-2+3\sqrt{3}}{4} \\ \frac{1+2\sqrt{3}}{2} \end{bmatrix}$$

The actual location is given by:

$$\mathbf{R}_x\left(\frac{\pi}{6}\right)\mathbf{R}_z\left(\frac{\pi}{3}\right)\mathbf{r} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{3}{4} & \frac{\sqrt{3}}{4} & -\frac{1}{2} \\ \frac{\sqrt{3}}{4} & \frac{1}{4} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{1-\sqrt{3}}{2} \\ \frac{-1+\sqrt{3}}{4} \\ \frac{1+5\sqrt{3}}{4} \end{bmatrix}$$

We can see that $\begin{bmatrix} \frac{-1+2\sqrt{3}}{4} \\ \frac{-2+3\sqrt{3}}{4} \\ \frac{1+2\sqrt{3}}{2} \end{bmatrix} \neq \begin{bmatrix} \frac{1-\sqrt{3}}{2} \\ \frac{-1+\sqrt{3}}{4} \\ \frac{1+5\sqrt{3}}{4} \end{bmatrix}$, and that they are not the same.

4. Braincells

After making your way through the last homework assignment, your remaining braincells are in disarray. Let's work together to get them back in order!

For all subparts of this problem, let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be an *adjacency matrix* representing the connections between each of your braincells with the other braincells. In particular, entry $\mathbf{B}_{ij}$ represents the speed of communication between braincell j and braincell i , specifically $j \rightarrow i$, where every $\mathbf{B}_{ij} = 0$ if there is no communication between braincell j and braincell i at all. Note that braincells do not communicate with themselves, so all diagonal entries should be 0.

- (a) At the very beginning, all of your braincells are disconnected from one another. Denote this adjacency matrix as $\mathbf{A}$. What kind of matrix is $\mathbf{A}$?

Solution:

Since all of our braincells are disconnected from one another, we can see that $\mathbf{A}$ will be a matrix of all zeros. As a result, $\mathbf{A}$ will be the zero matrix, $\mathbf{0}_{n \times n}$.

As the zero matrix, we can additionally note that $\mathbf{A}$ would simultaneously be upper triangular, lower triangular, and diagonal.

- (b) After a good night's rest, all of your brain connections are beginning to come back. In particular, the speed of communication between every pair of braincells has increased by 1. Assuming that the adjacency matrix before going to sleep was represented by $\mathbf{B}$, express this new adjacency matrix, $\mathbf{B}'$, in terms of $\mathbf{B}$ and constant matrices, as well as any scalar multiples.

Note: Recall in the problem statement that braincells do not communicate with themselves. Your new $\mathbf{B}'$ should preserve this property.

Solution:

Per the problem description, the speed of communication between every pair of braincells has increased by 1. Therefore, we would like to increment every entry of $\mathbf{B}$ by 1. This would be equivalent to the following:

$$\mathbf{B}'_{ij} = \mathbf{B}_{ij} + 1, \forall i, j \in \{1, 2, \dots, n\}$$

However, as the note reminds us, all of our diagonal entries should remain 0, since we assume braincells do not communicate with themselves. Therefore, we can rewrite our expression as follows:

$$\mathbf{B}'_{ij} = \begin{cases} \mathbf{B}_{ij} + 1 & \text{if } i \neq j, \\ \mathbf{B}_{ij} & \text{if } i = j, \end{cases} = \begin{cases} \mathbf{B}_{ij} + 1 & \text{if } i \neq j, \\ \mathbf{B}_{ij} + 1 - 1 & \text{if } i = j, \end{cases}$$

We can write this relationship in matrix form as:

$$\mathbf{B}' = \mathbf{B} + \mathbf{1}_{n \times n} - \mathbf{I}_n = \mathbf{B} + (\mathbf{1}_{n \times n} - \mathbf{I}_n)$$

- (c) After consuming your favorite source of caffeine, the communication speed between all of your braincells quadruples. Express this new adjacency matrix, $\mathbf{C}$, in terms of the adjacency matrix before caffeine, $\mathbf{B}$, and a constant matrix, as well as any scalar multiples.

Solution:

Per the problem description, the speed of communication between every pair of braincells has quadrupled. Therefore, we would like to scale every entry of $\mathbf{B}$ by 4. This is equivalent to the following:

$$\mathbf{C}_{ij} = 4\mathbf{B}_{ij}, \forall i, j \in \{1, 2, \dots, n\}$$

Written in matrix form, this means that:

$$\mathbf{C} = 4\mathbf{B}$$

- (d) All ready to go, you attend your favorite TA's discussion. After collaborating with your peers on the new course content, the communication speed between 2 braincells from k to braincell l increases by 10. The rest of the connections stay the same. Express the new adjacency matrix, $\mathbf{D}$, in terms of the adjacency matrix before discussion, $\mathbf{B}$, and a constant matrix, as well as any scalar multiples.

Note: If you cannot provide a closed-form mathematical expression for $\mathbf{D}$, a qualitative expression is also acceptable.

Solution:

In this case, the value of particular braincell connection is being incremented, and the rest stay the same. As a result, we can see that this is equivalent to:

$\mathbf{D} = \mathbf{B} + \mathbf{A}$, where $\mathbf{A}_{lk} = 10$ and all other entries of $\mathbf{A}$ are zero. In otherwords, we just increase the entry $\mathbf{B}_{lk}$ by 10 and the rest stay the same. Note that it is $\mathbf{B}_{lk}$ and not $\mathbf{B}_{kl}$ since the connection goes from $l \rightarrow k$.

- (e) For this subpart, imagine that the speed of communication between braincells i and j is equal in both directions. In other words, consider the case where $\mathbf{B}_{ij} = \mathbf{B}_{ji} \forall i, j \in \{1, 2, \dots, n\}$. In this case, what kind of matrix is $\mathbf{B}$?

Solution:

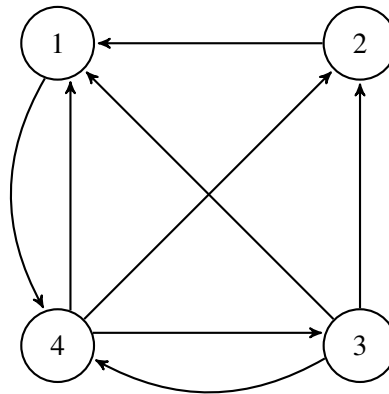
In this case, $\mathbf{B}$ is a symmetric matrix. We can use the following visualization to confirm this fact:

$$\mathbf{B}^\top = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} & \dots & \mathbf{B}_{1n} \\ \mathbf{B}_{21} & \mathbf{B}_{22} & \dots & \mathbf{B}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{B}_{n1} & \mathbf{B}_{n2} & \dots & \mathbf{B}_{nn} \end{bmatrix}^\top = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{21} & \dots & \mathbf{B}_{n1} \\ \mathbf{B}_{12} & \mathbf{B}_{22} & \dots & \mathbf{B}_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{B}_{1n} & \mathbf{B}_{2n} & \dots & \mathbf{B}_{nn} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} & \dots & \mathbf{B}_{1n} \\ \mathbf{B}_{21} & \mathbf{B}_{22} & \dots & \mathbf{B}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{B}_{n1} & \mathbf{B}_{n2} & \dots & \mathbf{B}_{nn} \end{bmatrix} = \mathbf{B}$$

5. Adjacency Matrix Walks

In this problem, we will use properties of adjacency matrices to learn about paths in graphs!

As a refresher, consider the following graph:



Graph A

In this graph, the circles with the numbers 1, 2, 3, 4 are **vertices**, and the arrows going between them are **edges**. For example, the arrow going from (3) to (1) is denoted as an edge, $3 \rightarrow 1$.

The definition of a **walk** between two vertices is a sequence of edges that starts at the first vertex and ends at the second vertex. These edges do not have to be unique; you can repeat edges. Since this is a **directed** graph, you can only go in the direction of the arrow. The **length** of this walk is the number of edges between the start and end vertices.

For example, in Graph A, the walk $4 \rightarrow 2 \rightarrow 1$ is a path between vertices 4 and 1 of length 2. However, the walk $1 \rightarrow 2 \rightarrow 4$ does not exist, as $1 \rightarrow 2$ and $2 \rightarrow 4$ are not edges in graph A.

For this problem, you are given an **arbitrary** graph G over n vertices, represented as an adjacency matrix $\mathbf{A}$ defined as follows:

$$\mathbf{A} = \{a_{ij} = 1 \text{ if the edge } j \rightarrow i \text{ exists in } G, \text{ else } 0\}$$

Assume you have no duplicate edges.

Show that in the matrix $\mathbf{B} = \mathbf{A}^2$, each element b_{ij} denotes the number of walks of length 2 from j to i in G .

Hint: Start by writing out b_{ij} explicitly.

Solution: In $\mathbf{B} = \mathbf{A}^2$, each element b_{ij} is defined by the following summation, based on the definition of matrix multiplication:

$$b_{ij} = \sum_{k=1}^n a_{ik}a_{kj}$$

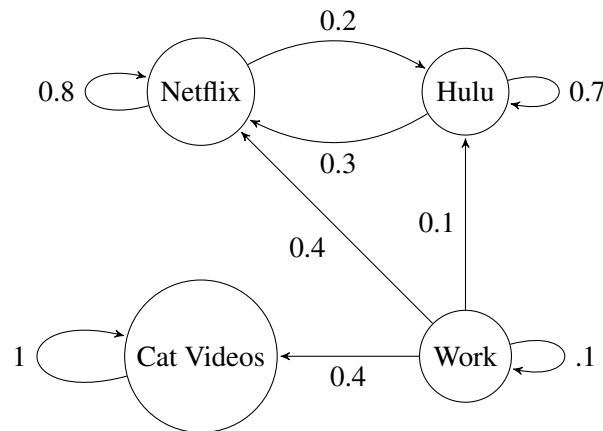
We can intuitively explain that $a_{ik}a_{kj}$ represents the edge from $j \rightarrow k$, if it exists, times the edge from $k \rightarrow i$, if it exists.

If either edge does not exist, one of the terms is 0, and the product is 0. Thus, we can see that this is only 1 if both the edges $j \rightarrow k$ and $k \rightarrow i$ exist, and so a walk $j \rightarrow k \rightarrow i$ must exist.

This is a walk of length 2 from j to i . By taking $\sum_{k=1}^n a_{ik}a_{kj}$, we are counting all walks from j to i in G that pass through *any* intermediate vertex k , not just one. Since all walks of length 2 need an intermediate vertex, b_{ij} will denote the number of walks of length 2 from j to i in G !

6. Social Media

As a tech-savvy Berkeley student, the distractions of streaming services are always calling you away from productive stuff like homework for your classes. You're curious—are you the only one who spends hours switching between Netflix or Hulu? How do other students manage to get stuff done and balance staying up to date with the Bachelor? You conduct an experiment, collect some data, and notice Berkeley students tend to follow a pattern of behavior similar to the figure below. So, for example, if $x = 100$ students are on Netflix, in the next timestep, 20 (i.e., $0.2 \cdot x$) of them will click on a link and move to Hulu, and 80 (i.e., $0.8 \cdot x$) will remain on Netflix.



This problem has an IPython component. Please attach a pdf printout of your IPython code to the HW in order to receive credit for problems that involve coding as stated in the submission format above.

- (a) Let us define $x_N[n]$ as the number of students on Netflix at time-step n ; $x_H[n]$ as the number of students on Hulu at time-step n ; $x_C[n]$ as the number of students watching any kind of cat video at time-step n ; and $x_W[n]$ as the number of students working at time-step n .

Let the state vector be: $\mathbf{x}[n] = \begin{bmatrix} x_N[n] \\ x_H[n] \\ x_C[n] \\ x_W[n] \end{bmatrix}$. Derive the corresponding transition matrix $\mathbf{A}$. You should

find that $\mathbf{A}$ is *block upper triangular*, with the form given below:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{A}_{22} \end{bmatrix},$$

where $\mathbf{A}_{11}, \mathbf{A}_{12}, \mathbf{A}_{22} \in \mathbb{R}^{2 \times 2}$ and the lower left block is the 2×2 zero matrix. Explain qualitatively why this is the case, referring to the given figure.

Hint: A transition matrix, $\mathbf{A}$, is the matrix that transitions $\mathbf{x}[n]$, the vector at time-step n to $\mathbf{x}[n+1]$, the vector at time-step $n+1$. In other words: $\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n]$.

Solution: Let us explicitly write the transition equation for each state and then we can use these to

identify the state transition matrix.

$$x_N[n+1] = 0.8 \cdot x_N[n] + 0.3 \cdot x_H[n] + 0.4 \cdot x_W[n]$$

$$x_H[n+1] = 0.2 \cdot x_N[n] + 0.7 \cdot x_H[n] + 0.1 \cdot x_W[n]$$

$$x_C[n+1] = 1 \cdot x_C[n] + 0.4 \cdot x_W[n]$$

$$x_W[n+1] = 0.1 \cdot x_W[n]$$

If $\mathbf{x}[n] = \begin{bmatrix} x_N[n] \\ x_H[n] \\ x_C[n] \\ x_W[n] \end{bmatrix}$, then for $\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n]$ we can identify the state transition matrix $\mathbf{A}$ as

$$\begin{bmatrix} 0.8 & 0.3 & 0 & 0.4 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0 & 1 & 0.4 \\ 0 & 0 & 0 & 0.1 \end{bmatrix}$$

- (b) There are 320 of you in the class. Suppose on a given Friday evening (the day when HW is due), there are 110 EECS16A students on Netflix, 60 on Hulu, 10 watching Cat Videos, and 140 actually doing work. In the next timestep, how many people will be doing each activity? In other words, after you apply the matrix once to reach the next timestep, what is the state vector?

Solution:

In order to calculate the state vector at the next timestep, we can use the equation $\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n]$. Substituting the values for $\mathbf{A}$ and $\mathbf{x}[n]$, we get the following:

$$\begin{bmatrix} 0.8 & 0.3 & 0 & 0.4 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0 & 1 & 0.4 \\ 0 & 0 & 0 & 0.1 \end{bmatrix} \begin{bmatrix} 110 \\ 60 \\ 10 \\ 140 \end{bmatrix} = \begin{bmatrix} 162 \\ 78 \\ 66 \\ 14 \end{bmatrix}$$

- (c) Compute the sum of each column in the state transition matrix. What is the interpretation of this?

Solution:

Since each column's sum is equal to 1, the system is conservative. This means that we aren't losing (or gaining) students after each time step and the total number of students remains constant.

- (d) Let $\mathbf{A}$ be the transition matrix you derived above. Complete your implementation in the provided Jupyter notebook.
- Implement a simulator that takes $\mathbf{A}$ and an initial state $\mathbf{x}[0]$, and iterates $\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n]$ until either $\|\mathbf{x}[n+1] - \mathbf{x}[n]\|_1 < 10^{-1}$ or $n = 1000$. Return all iterates $\mathbf{x}[0], \dots, \mathbf{x}[T^*]$ where T^* is the last index. For this and all parts after, **do not** round $\mathbf{x}$ elements to the nearest whole number.
 - Using your simulator with the initial state from part (a), $\mathbf{x}[0] = [110, 60, 10, 140]^T$:
 - Plot $x_N[n], x_H[n], x_C[n], x_W[n]$ versus n on one figure with labeled axes and a legend.
 - For each iterate, compute $S[n] = \mathbf{1}^T \mathbf{x}[n]$ and verify that $S[n] = 320$ for all n .
 - What is the steady-state vector $\hat{\mathbf{x}}$ you observe at termination and what are the proportions of students in each state?

Solution: The trajectories should show $x_W[n]$ decreasing toward 0 while the streaming states level out to about 149 (46.4%) students on Netflix, 99 (31.0%) students on Hulu, 72 (22.6%) students watching

cat videos, and 0 students working. The steady-state vector $\hat{\mathbf{x}}$ has all students on the streaming services and sums to 320. Also, $S[n]$ should stay constant at 320 for all n . See the solution notebook for an example of a correct implementation and plots.

- (e) We try a policy that increases the chance of staying in Work while keeping each column sum equal to 1. We do this by having students download a new app that makes students wait for a certain amount of time before they can start watching a streaming service. Note that once a student starts watching a streaming service, the app becomes inactive.

For every 1 minute that a student waits, the probability of staying in Work increases by 0.1. To help us decide how long to make students wait, we will vary the parameter α which represents the additional probability of staying in Work after the app is downloaded.

The original Work column of $\mathbf{A}$ is $\mathbf{w}^{(0)} = \begin{bmatrix} 0.4 \\ 0.1 \\ 0.4 \\ 0.1 \end{bmatrix}$. For a parameter $\alpha \in [0, 0.9]$, define the modified

Work column $\mathbf{w}(\alpha) : [0, 0.9] \rightarrow \mathbb{R}^4$ as

$$\mathbf{w}(\alpha) = \begin{bmatrix} 0.4 - \alpha \frac{0.4}{0.9} \\ 0.1 - \alpha \frac{0.1}{0.9} \\ 0.4 - \alpha \frac{0.4}{0.9} \\ 0.1 + \alpha \end{bmatrix},$$

and construct $\mathbf{A}(\alpha) : [0, 0.9] \rightarrow \mathbb{R}^{4 \times 4}$ by replacing only the Work column of $\mathbf{A}$ with $\mathbf{w}(\alpha)$ (all other columns unchanged).

- For $\alpha \in \{0.1, 0.2, \dots, 0.9\}$, simulate $\mathbf{x}[n+1] = \mathbf{A}(\alpha)\mathbf{x}[n]$ to convergence and plot $x_W[n]$ versus n for each α (with labeled axes and a legend).
- How long do we need to make students wait so that at least one student is still working in the steady state?

Solution:

- As α increases, $x_W[n]$ decays slower.
- For every $\alpha < 0.9$, numerical simulation should still show $x_W[n] \rightarrow 0$ eventually.
- Only for $\alpha = 0.9$ does the steady state have any students working (in fact, all students are working who started out working).
- See the solution notebook for an example of a correct implementation and plots.

- (f) Instead of an app that makes students wait, we ask students to download a different app that when a student is on a streaming service, blocks them from watching it for a certain amount of time. Note that this app only activates after a student has already started watching a streaming service. Each minute that a student is blocked, the probability of returning to Work from any streaming service increases by 0.1, while the probability of returning to the same streaming service decreases by 0.1. We can block students for up to 7 minutes.

Let γ be the number of minutes that a student is blocked. Let $\mathbf{A}(\gamma) : [0, 7] \rightarrow \mathbb{R}^{4 \times 4}$ be the transition matrix that results from blocking students for γ minutes.

- Derive an expression for $\mathbf{A}(\gamma)$ (i.e. write out the entries in terms of γ and the original entries of $\mathbf{A}$). Is this still a conservative system?
- Using your simulator from part (a), find the minimum γ such that, for the initial state $\mathbf{x}[0] = [110, 60, 10, 140]^\top$, at least half of the students who started out working are still working in the

steady state. You may round your answer to the nearest integer. For this γ , plot $x_N[n]$, $x_H[n]$, $x_C[n]$, $x_W[n]$ versus n on one figure with labeled axes and a legend.

Solution: For making the students wait γ minutes $\mathbf{A}(\gamma)$ is given by:

$$\mathbf{A}(\gamma) = \begin{bmatrix} 0.8 - \frac{\gamma}{10} & 0.3 & 0 & 0.4 \\ 0.2 & 0.7 - \frac{\gamma}{10} & 0 & 0.1 \\ 0 & 0 & 1 - \frac{\gamma}{10} & 0.4 \\ \frac{\gamma}{10} & \frac{\gamma}{10} & \frac{\gamma}{10} & 0.1 \end{bmatrix}$$

This is still a conservative system as each column sums to 1. You should find that $\gamma = 3$ is the minimum whole number of minutes that we need to block students so that at least half of the students who started out working are still working in the steady state. At 3 minutes of blocking, 80 students are still working in the steady state. Any number of minutes between 2.5 and 3 is an acceptable answer. See the solution notebook for an example of a correct simulation and plots.

7. Labeling Patients Using Gene Expression Data

Least squares techniques are useful for many different kinds of prediction problems. Numerous researchers have extensively further developed the core ideas that we have learned in class. These ideas are commonly used in machine learning for finance, healthcare, advertising, image processing, and many other fields. Here, we'll explore how least squares can be used for classification of data in a medical context.

Gene expression data of patients, along with other factors such as height, weight, age, and family history, are often used to predict the likelihood that a patient might develop a certain disease. This data can be combined into a vector that describes each patient. This vector is often referred to as a feature vector.

Many scientific studies examine mice to understand how gene expression relates to diabetes in humans. Studies have shown that the expression of the *tomosinB* and *tsA* genes are correlated to the onset of diabetes in mice. How can we predict whether or not a mouse will develop diabetes based on data about this expression as well as other factors of the mouse? We will use some (fake) data to explore this.

We are given feature vectors for each mouse as:

$$\begin{bmatrix} \text{age} \\ \text{weight} \\ \text{tomosinB} \\ \text{tsA} \\ \text{chnA} \end{bmatrix}$$

Age and weight in the vector above are represented by real numbers, while the presence or absence of the expression of the genes *tomosinB*, *tsA*, and *chnA* is captured by $+1$ and -1 respectively. For example, the vector $[2 \ 20 \ 1 \ -1 \ -1]^T$ means a 2 month old mouse, that weighs 20 grams, expresses the genes *tomosinB*, but not *tsA* or *chnA*.

We would like the following expression to be **positive if the mouse has diabetes and negative if the mouse does not have diabetes**:

$$f(\text{age}, \text{weight}, \text{tomosinB}, \text{tsA}, \text{chnA}) = \alpha_1(\text{age}) + \alpha_2(\text{weight}) + \alpha_3(\text{tomosinB}) + \alpha_4(\text{tsA}) + \alpha_5(\text{chnA}). \quad (3)$$

- (a) We wish to set up a linear model for the problem in the format $\mathbf{Ax} = \mathbf{b}$. Here, $\mathbf{b}$ will be a vector with $+1, -1$ entries where a 1 represents that the mouse is diabetic and -1 represents that the mouse is not diabetic. The feature vectors of each mouse will be included in each row of the matrix $\mathbf{A}$. For example, if the first row of $\mathbf{A}$ contains the data of mouse #1 and the first entry of $\mathbf{b}$ is $+1$, that means the mouse #1 has diabetes.

Set up the problem by writing $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$ in terms of the variables in the feature vectors, α_i , and any other variables you define. What are your unknowns?

Solution: Assume we have n mice. Define b_j as the indicator for whether the j -th mouse has diabetes. Each of the data variables will be indexed by $j = 1, \dots, n$ for each of the n mice. The unknowns that we want to find are α_i , where $\{i = 1, \dots, 5\}$. The vector $\mathbf{x}$ is the vector of unknowns $\mathbf{x} = [\alpha_1 \ \alpha_2 \ \alpha_3 \ \alpha_4 \ \alpha_5]^T$. The complete problem setup is:

$$\underbrace{\begin{bmatrix} \text{age}_1 & \text{weight}_1 & \text{tomosinB}_1 & \text{tsA}_1 & \text{chnA}_1 \\ \text{age}_2 & \text{weight}_2 & \text{tomosinB}_2 & \text{tsA}_2 & \text{chnA}_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \text{age}_n & \text{weight}_n & \text{tomosinB}_n & \text{tsA}_n & \text{chnA}_n \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_{\mathbf{b}}$$

- (b) Training data is data that is used to develop your model. The matrix $\mathbf{A}$ and vector $\mathbf{b}$, provided in `gene_data_train.npy` and `diabetes_train.npy` respectively, represent the (fake) *training* data. Use the data to find the optimal model parameters $\alpha_1, \dots, \alpha_5$ for the given data set. Find the optimal parameter values using least squares method and the provided IPython notebook.

Solution:

To solve for $\mathbf{x}$ in $\mathbf{Ax} = \mathbf{b}$ using the least squares technique, we find $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$. The result is

$$\mathbf{x} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = \begin{bmatrix} 0.12131475 \\ -0.15253102 \\ -0.36111722 \\ -0.06427341 \\ 0.95936096 \end{bmatrix}$$

- (c) Now it is time to use the model you have developed to make some predictions! It is interesting to note here that we are not looking for a real number to model whether each mouse has diabetes or not; we are looking for a **binary label**. Therefore, we will use the **sign of the expression from Equation (3)** to assign a ± 1 value to each mouse. Each mouse characteristics are represented by each row of `gene_data_test.npy`.

Predict whether each mouse with the characteristics in the *test* data set `gene_data_test.npy` will get diabetes. There are four mice/ rows in the test data set. Calculate the ± 1 prediction vector $\mathbf{b}_{calc}$ that shows the prediction whether a mouse will be diabetic or not. Observe the ± 1 vector $\mathbf{b}_{test}$ from `diabetes_test.npy` that indicates whether or not the mice *actually* have diabetes.

What is the prediction accuracy (number of correct predictions divided by total number of predictions) of your model?

Solution:

Using the values of α_i calculated in the previous part on the test data, we see that the prediction is $\mathbf{b} = [1 \ 1 \ 1 \ 1]^T$, which differs in the third entry of the given file `diabetes_test.npy`. Therefore the prediction accuracy is 75%.

8. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.